



Combined Heat and Power Producing Company Limited

Term of Reference

Development of a Centralized Platform for the 24-MW Solar Project of Combined Heat and Power Producing Co., Ltd. (CHPP) under a Power Purchase Agreement (PPA)

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อนุมัติโดย	
ชื่อ-สกุล	
ตำแหน่ง	



เรื่อง: งานจ้างพัฒนาแพลตฟอร์มศูนย์กลางสำหรับโครงการโซลาร์ของบริษัท ผลิตไฟฟ้าและพลังงานร่วม จำกัด (CHPP) ขนาด 24 MW (สัญญา PPA)		
จัดทำโดย: ฝ่ายวิศวกรรมและโครงการก่อสร้าง	วันที่จัดทำ: 12/12/2568	อ้างอิง (PR) เลขที่ XXXXXXXXXX

List of contents

1. Procurement Title	4
2. Procurement Objectives	4
3. Shipping Location / Project Location	4
4. Proposer qualifications	4
5. General guidelines for submitting proposal	4
6. Proposal documents/Price confirmation schedule	5
7. Bid Security	5
8. Criteria for selecting bidders	5
9. Scope of Work	6
10. Material properties requirements	10
11. General Requirements	10
12. Project management	10
13. Delivery Period	10
14. Increasing or Decreasing the scope of service	10
15. Utilities and other facilities	10
16. Working Conditions/Work Specific Conditions	10
17. Conditions upon termination	10
18. Work inspection and acceptance	11
19. Payment terms	11
20. Billing Documents / Payment Period	11
21. Billing Documents / Payment Period	12



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อ้างอิง (PR) เลขที่ XXXXXXXXXX

22. Performance Guarantee	12
23. Performance Bond / Guarantee Bond	13
24. Safety, health and environmental requirements	13
25. Other conditions	13
26. Penalty/Compensation for damages	13
27. Contract draft	14
28. Attachment	14



เรื่อง: งานจ้างพัฒนาแพลตฟอร์มศูนย์กลางสำหรับโครงการโซลาร์ของบริษัท ผลิตไฟฟ้าและพลังงานร่วม จำกัด (CHPP) ขนาด 24 MW (สัญญา PPA)		
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1. Procurement Title

Development of a Centralized Platform for the 24-MW Solar Project of Combined Heat and Power Producing Co., Ltd. (CHPP) under a Power Purchase Agreement (PPA)

2. Procurement Objectives

The Combined Heat and Power Producing Company Limited (CHPP) intends to procure the development and deployment of a Centralized Platform for its 24 -MW Solar Project under a Power Purchase Agreement (PPA). This Terms of Reference (TOR) defines the required features, functional specifications, and technical capabilities for implementing BaxEnergy Farsight with enhanced Solar Power Management functions, serving as a unified system for real-time monitoring, performance analysis, reporting, and operational oversight of CHPP's solar assets. The TOR provides sufficient detail to enable bidders to prepare complete and competitive proposals for the successful implementation of the Platform.

3. Shipping Location / Project Location

Combined Heat and Power Producing Co., Ltd

Government Center 120 Moo 3, Energy Building, Government Complex Commemorating His Majesty's 80th Birthday, December 5, 2007, Chaeng Watthana Road, Thung Song Hong, Lak Si, Bangkok 10210.

4. Proposer qualifications

4.1 The proposer must be a corporation registered with the Ministry of Commerce.

4.2 The proposer must have a registered capital of not less than 1,000,000 baht

4.3 The proposer must have experience as a legal advisor

5. General guidelines for submitting proposal

5.1 The proposer must study the details in the specification documents thoroughly before submitting the proposal to understand the scope of work and the clarity of the work in order to avoid mistakes.

5.2 The proposer agrees to be responsible for paying stamp duty (if any) and any other taxes (if any) related to



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6. Proposal documents/Price confirmation schedule

6.1 Supporting Documents for the Proposal

6.1.1 Technical Proposal

6.1.1.1 Description of proposed scope of service

6.1.1.2 Proposed execution plan, schedule, and timeframe

6.2 Price Validity

The Bidder shall provide a price validity period of not less than 30 days from the date of proposal submission, or as otherwise specified by the Employer. During the validity period, the Bidder shall be fully responsible for the proposed price and shall not revise the price and/or withdraw the proposal under any circumstances.

7. Bid Security

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8. Criteria for selecting bidders

8.1 The technical proposals will be opened and reviewed to determine their acceptability and responsiveness to the specifications. Unacceptable and non responsive bids shall be rejected and the commercial proposals will be retaining unopened. Correspondence will be held with the accepted Bidders as necessary to bring their bids to acceptable technical standard

8.2 The notice will also be sent to the Bidders whose bids have been rejected, upon receipt of such notice, the Bidders may collect from CHPP the unopened commercial proposals within thirty (30) days, after this period any proposals of rejected bids retained by CHPP shall be destroyed and unread.

8.3 CHPP will not be bound to accept the bid with the lowest submitted price. CHPP reserves the right to accept the bid, which is evaluated by CHPP as the proper bid.

8.4 CHPP will award the Contract to the successful Bidder, whose bid has been determined to be substantially responsive and has been determined as the proper evaluated bid, provided further, that the Bidder is determined as being qualified to perform the Contract satisfactorily.



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อ้างอิง (PR) เลขที่ XXXXXXXXXX

8.5 CHPP reserves the right to accept or reject any bid, and to annul the bidding process and reject all bids at any time without an obligation to inform the affected bidder or bidders of the grounds for CHPP's decision prior to award the Contract, without any liabilities to the affected Bidder or Bidders.

9. Scope of Work

The Scope of Work includes deploying, configuring, and commissioning the Centralized Solar Management Platform for CHPP. This also covers integrating the system with all designated solar sites. The Contractor will implement the software, migrate and validate data, establish real-time connectivity, and configure all required functions to support monitoring, analysis, and operational management. Subsequent sections specify detailed functional capabilities, including but not limited to the following:

9.1 OFFLINE / IN-CONSTRUCTION PLANT SUPPORT

- Ability to register, display, and track plants that are under construction, commissioned, or temporarily offline

9.2 PORTFOLIO MONITORING

- Unified dashboard presenting all solar plants in the portfolio.
- Summary KPIs such as total capacity, production, availability, alarms, and health status.
- Quick comparison across sites for performance benchmarking.

9.3 ASSET-TYPE FILTERING

- Filters to categorize and display assets based on type: Solar plants, wind farms, inverters.

9.4 DRILLDOWN FROM PORTFOLIO TO ASSET/DEVICE LEVEL

- Intuitive navigation from portfolio level to individual plants.
- Further drilldown to device categories (e.g. inverters) and individual device details.
- Detailed views including status, alarms, KPI, and performance charts.

9.5 POWER PLANT SCADA SCREENS

- Real-time SCADA mimic diagrams for each plant.
- Live status indicators for monitoring operations and alarms.

9.6 SOLAR PLANT KEY PARAMETER MONITORING

- Tracking of essential parameters such as irradiance, active power, Availability and production (MWh).



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อ้างอิง (PR) เลขที่ XXXXXXXXXX

9.7 INVERTER LIVE TRENDS

- Live trend charts for inverter parameters such as model, power in string level, Active Power.

9.8 LIVE ALARMS BY CATEGORY

- Real-time alarm dashboard display.
- Clear visual differentiation for critical, major, minor, and warning alarms

9.9 ALARM FILTERING & SORTING

- Flexible filtering by: Alarm name, Site, Area, SCADA, Device, Message, Signal type, Device type, Severity
- Sorting by timestamp, severity, asset name, or duration.

9.10 ALARM ACKNOWLEDGMENT

- User can acknowledge alarm directly through the interface.

9.11 ALARM LOG & HISTORY

- Complete historical database of alarms.

9.12 ADVANCED ALARM CONFIGURATOR

- Configurator for custom alarm rules and logic.
- Support for conditional triggers based on criteria.

9.13 TIME-SERIES TREND ANALYSIS

- Time-series analysis for all major parameters (reactive power, active power, solar irradiance, energy exported, frequency, efficiency).

9.14 INTERACTIVE CHART (ZOOM/PAN)

- Fully interactive charting tools enabling zoom-in, zoom-out and panning.
- Comparison of multiple parameters or devices on a single timeline.

9.15 XY SCATTER PLOT

- Scatter plot visualization for performance analytics.
- Useful for identifying performance deviations or equipment degradation.

9.16 SOLAR POWER CURVE ANALYSIS

- Detection of underperformance, clipping, or weather-related deviations.

9.17 EVENT LOGBOOK (FAULT, MAINTENANCE, WARNING)



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- Comprehensive event timeline capturing all events including faults, warnings and maintenance actions.

9.18 EXPORT FUNCTIONALITY (.XLSX FORMAT)

- One-click export of reports in .xlsx format.

9.19 PERSONALIZED DRAG & DROP DASHBOARDS

- User-customizable dashboards with drag and drop widgets.
- Support for multiple dashboard layouts per user role or purpose.

9.20 CROSS DEVICE COMPATIBILITY

- Responsive UI optimized for desktops, tablets, and mobile devices.
- Consistent functionality across different screen sizes and browsers.

9.21 FAULT AND MAINTENANCE REPORT

- Daily, weekly, or monthly report summarizing faults, maintenance actions, and downtime trends.
- Insights for reliability, root-cause analysis, and operational planning.

9.22 GIS MAP WITH REAL-TIME ASSET STATUS

- Geographic Information System (GIS) map showing all solar sites with real time statuses.
- Layered views for availability, production, and weather.

9.23 DRILLDOWN IN MAP VIEW (REGION/SITE/DEVICE)

- Interactive map drilldown support navigation from country/region and site.
- Map-based selection and monitoring.

9.24 EXECUTIVE PORTFOLIO REPORT

- High level summary report including KPIs such as production, target production, lost production, revenue, energy loss, availability.
- Suitable for management-level weekly or monthly reviews.

9.25 REPORT SCHEDULER

- Automated report generation and distribution via email.



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วันที่จัดทำ: 12/12/2568

อ้างอิง (PR) เลขที่ XXXXXXXXXX

9.26 DYNAMIC ASSET GROUP CREATION

- Users can create custom asset groups based on criteria such as cut in, cut out, latitude, longitude, Rotor Diameter.

9.27 CONNECTION DEVICES

- The platform shall connect with multi-vendor Energy meter with API standard protocol or Modbus protocol.

The Platform shall integrate and process data from all designated solar sites under the CHPP Solar Project. The details of each site, including installed capacity, location, and system configuration, are provided in the table below.

NO	Power Plant Name	Nominal Power [kW]	City/Town
1	Solar1 (CHPP Solar Farm)	5,000	Chanthaburi
2	VISTEC I	500	Rayong
3	PTTOR Talingchan	48	Bangkok
4	PTTOR Minburi	127	Bangkok
5	KVIS I	100	Rayong
6	THAIOIL I	500	Chonburi
7	VISTEC II	890	Rayong
8	THAIOIL II	500	Chonburi
9	Future Flex	904	Samutsakorn
10	SUT	6,000	Nakornrachsim
11	PTTEP PTIC	238	Rayong
12	PTTEP RASC	282	Rayong
13	AICA	3,800	Ayutthaya
14	INI	1,600	Ayutthaya
15	Enter	548	Bangkok
16	SURETEX	3,000	Surathani
17	KVIS II	88	Rayong



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อ้างอิง (PR) เลขที่ XXXXXXXXXX

10. Material properties requirements

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11. General Requirements

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12. Project management

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13. Delivery Period

Delivery shall be completed within the required timeframe and no later than December 26, 2025 following receipt of the Purchase Order (PO). The Contractor shall deliver the software fully operational by the specified date, after which the Contractor must complete data integration for all designated sites.

14. Increasing or Decreasing the scope of service

The Employer reserves the right to amend, modify, or reduce the scope of services from the agreed-upon specifications and details without terminating this engagement. In the event that any additional requirements arise beyond those specified in the Terms of Reference, the Bidder agrees that such additions shall be documented in writing and duly signed by both parties.

15. Utilities and other facilities

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16. Working Conditions/Work Specific Conditions

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17. Conditions upon termination

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อ้างอิง (PR) เลขที่ XXXXXXXXXX

18. Work inspection and acceptance

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19. Payment terms

Payments shall be made in two (2) installments as follows:

19.1 The first installment shall be 100% of the Software License Fee, payable upon successful delivery of the software.

19.2 The second installment shall be 100% of the Engineer Fee for initial setup and first-year implementation, payable upon completion of all contracted works.

20. Billing Documents / Payment Period

20.1 Required Documents for Billing Submission

20.1.1 Invoice

20.1.2 Receipt or Payment Acknowledgment, with a copy of the national ID card and signature (for non-juristic persons), and/or Tax Invoice

20.1.3 Copy of the Purchase Order, Employment Confirmation Letter, or the fully executed Contract signed by both parties

20.1.4 Copy of the Work Notification Form (if applicable)

20.1.5 Goods Receipt / Work Delivery Note, or Certificate of Completion / E-mail confirmation of work acceptance

20.1.6 Reply e-mail from the CHPP Warehouse Team (if applicable)

20.2 Address for Receipt / Tax Invoice /Invoice

Combined Heat and Power Producing Company Limited

Tax Identification Number 0105550089831, Head Office

Energy Complex, Building B, 5th Floor 555/2 Vibhavadi Rangsit Road, Chatuchak , Chatuchak , Bangkok 10900

20.3 E-mail for sending documents for work acceptance

invoice@chpp.co.th



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20.4 Billing Place

Bangkok Province Location: Pick& Place Center

900/31, SV City Building, No. 12, LG Floor, Room No. 4/2-6, Phra Ram 3 Road,

Bang Phong Phang , Yan Nawa District, Bangkok 10120

Business Hours: Every business day, 8:30 a.m. to 4:00 p.m

21. Billing Documents / Payment Period

Payment shall be made by cheque, by paying at the counter of Standard Chartered Bank (Thai) Public Company Limited within 30 days from the date on which the counterparty has submitted the documents under Section 20.1 to Standard Chartered Bank (Thai) Public Company Limited in full.

Advance Payment / Payment (if have) will be made by bank transfer within 15 days from the date the vendor submits the documents according to Section 20.1 to the Accounting Department, ENCO B in full.

Contact Khun Punyawee, contact Khun 02- 140-4600 (Accounting Department)

22. Performance Guarantee

The Contractor shall ensure that the software platform and all subscribed monitoring services perform in accordance with the agreed service levels throughout the subscription period. The Contractor shall maintain system availability, data accuracy, real-time data acquisition, and functional performance as specified in the Service Level Agreement (SLA).

In the event that the system performance does not meet the required standards, CHPP reserves the right to request corrective action, service credit, or other remedies as specified in the Contract. The Contractor shall provide continuous technical support and maintenance for the duration of the annual subscription period to ensure uninterrupted system operation.



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23. Performance Bond / Guarantee Bond

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24. Safety, health and environmental requirements

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25. Other conditions

Confidentiality

- All information obtained by the consultant from CHPP and CHPP's affiliates during negotiation or throughout the services, all records or documents from CHPP and CHPP's affiliates pertaining to services, as well as the final results of services and items which are marked "Confidential" shall be treated with strictest confidentiality. The consultant must not, without prior written consent from CHPP, disclose to a third party or publish such information, records, documents or results (more conditions of confidential information and confidential obligations will be added upon effective Engagement Letter/NDA). This confidential clause is binding and shall be governed by the law of Thailand.

26. Penalty/Compensation for damages

During the execution period and warranty period, if the Contractor fails to complete the work within the specified timeframe, the following penalties shall apply:

26.1 A penalty rate of 0.1% per day for service-based installation work.

26.2 A penalty rate of 0.2% per day for procurement-based work.

26.3 The applicable penalty shall be calculated based on either Clause 26.1 or 26.2, depending on the nature of the work. Any fraction of a day shall be counted as a full day. The total penalty shall not exceed 10% of the total contract value. If the delay is not caused by the Contractor, the Contractor shall submit a written request to the Employer for an extension of the contract period. The Employer will consider and approve such requests on a case-by-case basis, and shall not deny an extension if the Contractor has justifiable grounds.



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27. Contract draft

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28. Attachment

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